

Carlos Aguilar

Email | Phone | GitHub | LinkedIn

EDUCATION

Hunter College (CUNY)

Bachelor of Arts in Computer Science, Minor in Mathematics

Expected Graduation: May 2027

GPA: 3.3/4.0

Relevant Coursework: Intro to Computer Science, Software Analysis & Design I–III, Computer Architecture, Intro to Data Science, Discrete Mathematics, Calculus I & II, Linear Algebra, Applied Statistics

TECHNICAL SKILLS

Programming Languages: Python, C++, Swift, JavaScript, HTML, CSS

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Scikit-Learn

Tools: Git, GitHub, Linux, VS Code, Jupyter Notebook, Microsoft Excel

Concepts: Data Structures, Algorithms, Object-Oriented Programming, Statistical Analysis, Software Design

PROJECTS

Spatial Registry Using Binary Search Trees | C++

- Designed and implemented a Binary Search Tree (BST) for organizing and querying 3D spatial data.
- Developed a Vector3D class to represent points and directions in three-dimensional space.
- Implemented nearest-neighbor and distance-based range query functionality using Euclidean magnitude calculations.
- Applied object-oriented programming, inheritance, linked data structures, and algorithmic problem-solving principles.

AI Application Project | Swift, Python, C++

- Currently developing an AI-driven application integrating multiple programming languages and technologies.
- Building functionality for data processing, model integration, and user interaction.
- Collaborating across software design, implementation, and testing phases.

EXPERIENCE

Programming Tutor

- Assisted students with programming concepts, debugging, and problem-solving strategies.

Mathematics Tutor

- Tutored students in algebra, calculus, linear algebra, and statistics.

English Language Tutor

- Helped students improve reading comprehension, writing, and communication skills.

LEADERSHIP & ACTIVITIES

- Member, Computer Science Club
- Member, Korean Student Association
- Interested in hackathons, software engineering, data science, and quantitative finance

HONORS & AWARDS

- TheDream.US Scholarship Recipient
- Graduated High School One Year Early
- Principal's Honor Roll (Multiple Years)